
FSS WORKING PAPER

SENTIMENT AND HOUSING PRICES

WRITTEN BY

DONG-JIN PYO

Financial Supervision Research Center

JUNGHO KIM

Banking Risk Office

THE VIEWS AND OPINIONS EXPRESSED HEREIN ARE SOLELY THOSE OF AUTHORS AND DO NOT REPRESENT THE VIEWS OF THE FSS.
--

2018.04

FINANCIAL SUPERVISORY SERVICE

Contents

1	Introduction	2
2	Model	3
3	Data	6
4	Results	7
5	Concluding Remarks	9
A	Sentiment Analysis	13

List of Figures

1	Monthly Sentiment Index (SI) and Housing Price Indices (HPI)	7
2	Daily Sentiment Index	8
3	Sentiment Volatility	8
4	The Process of Sentiment Analysis	13

List of Tables

1	Descriptive Statistics	7
2	Correlations between Sentiment Index and Housing Price Index	8
3	Correlations between Sentiment Volatility and Housing Price Volatility	8
4	Granger-Causality Tests: Nationwide	9
5	Granger-Causality Tests: Seoul	9
6	Seemingly Unrelated Regression Estimates	10

Sentiment and Housing Prices ^{*}

Dong-Jin Pyo[†]

Jungho Kim[‡]

Abstract

This paper quantifies the Korean housing market sentiment and investigates its statistical relationships with future housing prices and volatility. The mood of the Korean housing market is inferred from housing market-related news articles via text-mining techniques. The authors find the sentiment index Granger-causes a month-ahead housing price change, but not vice versa. For the nationwide housing price index, it is shown an increase in sentiment predicts a rise in housing price and a drop in volatility. However, sentiment volatility does not have a statistically significant impact. For the Seoul metropolitan area, the authors show sentiment is positively associated with the month-ahead housing price; whereas, an increase in sentiment volatility depresses housing prices.

Keywords: sentiment, housing price, volatility, news, text-mining, sentiment analysis

JEL classification: G12, R30

^{*}The views and opinions expressed here do not represent those of the institution with which authors are affiliated.

[†]First and corresponding author. Financial Supervision Research Center, Financial Supervisory Service. Email: djpyo@fss.or.kr. Address: 38 Yeoui-Daero, Yeongdeungpo-Gu, Seoul 07321, KOREA. Tel: +82-2-3145-8601.

[‡]Banking Risk Office, Financial Supervisory Service. Email: arak0806@fss.or.kr. Address: 38 Yeoui-Daero, Yeongdeungpo-Gu, Seoul 07321, KOREA. Tel: +82-2-3145-8358.

Housing has also been a source of endless fascination for the general public, because we live in houses, we work on them every day, and our sense of our individual social position is tied to the kind of houses we live in.

- Shiller (2005)

1. Introduction

In this paper, the housing market sentiment index is constructed from online news articles, a type of big data,¹ to examine its relationships with key variables in the Korean housing market. In particular, this study investigates whether the tone of unstructured qualitative data could provide valuable information on future housing market conditions.

Sentiment or psychology in the real estate market can play a critical role in shaping market dynamics because, in the real estate market, unlike financial markets, such as stocks, the tools (e.g., short selling) for adjusting housing prices, even if the price deviates from the fundamentals, is relatively limited.² Households often fail to make rational choices due to psychological biases. As a result, real-estate markets operating with anticipation for the future may show drastic changes due to extreme optimism or pessimism. The importance of the real estate market in South Korea is also becoming paramount because real estate assets account for the largest portion of household assets, and mortgage debts are tightly connected to the stability of the banking system.

Since the seminal paper by Baker and Wurgler (2006), numerous studies have attempted to quantify market sentiment or mood using either quantitative or qualitative sources. The former typically quantifies this sentiment through the lens of household surveys, while the latter relies on various textual data, including newspaper articles, annual reports, regulatory filings, etc. However, previous studies have largely focused on the financial market application,³ leaving the housing market relatively untapped.

It is surprising only a small number of studies undertake the housing market through the framework of sentiment based on qualitative data, provided sentiment (either optimism or pessimism) can possibly have a significant impact on housing prices because of high transaction costs and heterogeneity in housing goods.⁴ Soo (2015) is one of a few studies that attempts to quantify the housing

¹In contrast with rapid expansion of big data and its related technologies, economic analysis using big data in the economics literature and policy-making arena is an early stage and very limited. Bholat et al. (2015) notes, although the potential for Big Data is large, the utilization of it by central banks or regulators is very low. Nyman-Andersen (2016) also argues the usefulness of Big Data in the process of policy formation. A large number of policy-makers and research institutes analyze and forecast the markets, relying on structured data in the form of numbers. Many analytical models are based on low frequency data, which yield a structural limitation to capture the risks contained in the system in a prompt manner. Low-frequency data are disadvantageous in the sense that timeliness is degraded because it is disclosed with significant time lags.

²In traditional economic and financial theories, the view is that that economics agents are rational (e.g., expected utility maximization), therefore, the market is efficient. However, in the field of behavioral economics and experimental economics, there is a continuing discovery of empirical evidence against the assumptions underlying these traditional economic theories.

³See Antweiler and Frank (2004), Doran et al. (2010), Garcia (2013), Pyo and Kim (2017), Kothari et al. (2009), Li (2010), Tetlock (2007), and Tetlock et al. (2008).

⁴Piazzesi and Schneider (2009) show, in a simple theoretical setting, a small number of optimistic households can cause

market sentiment. She built the sentiment index for 34 cities across the U.S. from news articles published in local newspapers, showing the sentiment index from the tone of news articles has a predictive power for future housing prices.

We take a similar approach, but with a more advanced extraction technique. Soo (2015) constructs the sentiment index based on the frequency of positive or negative words preclassified according to the Harvard IV-4 dictionary. However, this paper only uses a small set of seed words to construct our own sentiment dictionary to base the semantic orientation for each selected word used to evaluate the news articles.

To our best knowledge, this study is the first endeavor to utilize the text-mining approach for Korean housing market analysis. Our findings suggest the text-based sentiment index is closely related to future housing prices and volatility. In particular, a rise in the sentiment index leads to an increase in the month-ahead housing price. On the other hand, we find, as the tone of news article becomes more optimistic, the volatility of housing prices wanes.

In Section 2, we consider a simple analytical model from which an econometric model is derived. In Section 3, we provide a brief description on data used for empirical analysis. In Section 4, we report key results of the estimation and discuss them.

2. Model

In this section, we present a simple model where the econometric specification of empirical model is derived. The model we are proposing here is a modified version of Danielsson et al. (2010) in which we introduce a sentiment element into the household expectation on housing price change. In the model, this stochastic sentiment acts as a unique seed of uncertainty to drive the evolution of equilibrium housing price change and its volatility.

Consider a representative household endowed with his own capital (K_t). Assume the purchase of houses is partially financed by a mortgage loan (B_t). The budget constraint for this household is given by

$$K_t + B_t = P_t \theta_t \quad (1)$$

where P_t is the market price of one house and θ_t denotes the demand for houses.

In a continuous time fashion, the capital gain (dK_t) can be written as

$$dK_t = dP_t \theta_t - dB_t \quad (2)$$

Assume the equilibrium housing price diffusion follows a geometric Brownian motion.

$$dP_t/P_t = \mu_t dt + \sigma_t dW_t \quad (3)$$

a dramatic change in housing prices.

where μ_t is the rate of the housing price change, σ_t denotes the volatility of the housing price change, and W_t follows a standard Wiener process.

The household is assumed to have a stochastic sentiment (ϵ_t) in the subjective perception on the true return of housing prices (μ_t), such that

$$x_t = \mu_t + \epsilon_t \quad (4)$$

where $d\epsilon_t = \sigma_\epsilon dW_t$. Note, σ_ϵ captures the volatility of the household's sentiment, which we label 'sentiment volatility.'

Conditional on the subjective belief (x_t), the expected capital gain by purchasing houses is given by

$$E(dK_t|x_t) = P_t\theta_t(x_t - r) + rK_t \quad (5)$$

Given the equilibrium volatility of housing price change, the variance of the household's capital gain is

$$Var(dK_t) = (P_t\theta_t\sigma_t)^2 \quad (6)$$

The household is assumed to maximize its mean-variance utility over capital gain from the investment on housing properties;

$$\begin{aligned} \max_{\theta_t} U &= E(dK_t|y_{it}) - \frac{1}{2}\lambda Var(dK_t) \\ &= [P_t\theta_t(x_t - r) + rK_t] - \frac{1}{2}(P_t\theta_t\sigma_t)^2 \\ s.t. \\ \theta_t &\geq 0 \end{aligned} \quad (7)$$

where λ denotes the coefficient of risk aversion. The first order condition for optimality yields the following demand function;

$$\theta_t(P_t; x_t) = \frac{(x_t - r)}{\lambda P_t \sigma_t^2} \quad (8)$$

Assuming the fixed supply of housing properties ($\bar{\theta}$), the equilibrium price of a house can be expressed as:

$$P_t = \frac{(x_t - r)}{\lambda \bar{\theta} \sigma_t^2} = \frac{(\mu_t + \epsilon_t - r)}{\lambda \bar{\theta} \sigma_t^2} \quad (9)$$

Note, from Eq. (9) we can infer

$$d\ln P_t = d\ln(\mu_t + \epsilon_t - r) - 2d\ln \sigma_t \quad (10)$$

Applying Ito's Lemma to the first term in Eq. (10), we have

$$\begin{aligned}
d\ln(\mu_t + \epsilon_t - r) &= \left(\frac{1}{\mu_t + \epsilon_t - r}\right)(\mu'_t + 1)d\epsilon_t + \frac{1}{2}\left(\frac{1}{\mu_t + \epsilon_t - r}\right)[\mu''_t - \left(\frac{1}{\mu_t + \epsilon_t - r}\right)(\mu'_t + 1)](d\epsilon_t)^2 \\
&= \frac{1}{2}\left(\frac{1}{\mu_t + \epsilon_t - r}\right)[\mu''_t - \left(\frac{1}{\mu_t + \epsilon_t - r}\right)(\mu'_t + 1)]dt + \left(\frac{1}{\mu_t + \epsilon_t - r}\right)(\mu'_t + 1)dW_t
\end{aligned} \tag{11}$$

where $\mu'_t = \frac{\partial \mu_t}{\partial \epsilon_t}$ and $\mu''_t = \frac{\partial^2 \mu_t}{\partial \epsilon_t^2}$. In a similar fashion, it follows

$$\begin{aligned}
d\ln\sigma_t &= \frac{\sigma'_t}{\sigma_t}d\epsilon_t - \frac{1}{2}\left(-\frac{\sigma'_t}{\sigma_t^2} + \frac{\sigma''_t}{\sigma_t}\right)(d\epsilon_t)^2 \\
&= \frac{1}{2}\left(\frac{\sigma'_t}{\sigma_t} - \frac{\sigma'_t}{\sigma_t^2}\right)dt + \frac{\sigma'_t}{\sigma_t}dW_t
\end{aligned} \tag{12}$$

where $\sigma'_t = \frac{\partial \sigma_t}{\partial \epsilon_t}$ and $\sigma''_t = \frac{\partial^2 \sigma_t}{\partial \epsilon_t^2}$. By Eq. (9), Eq. (11), and Eq. (12), we have

$$\begin{aligned}
d(\ln P_t) &= \left[\frac{1}{2}\left(\frac{1}{\mu_t + \epsilon_t - r}\right)[\mu''_t - \left(\frac{1}{\mu_t + \epsilon_t - r}\right)(\mu'_t + 1)] + \frac{1}{2}\left(\frac{\sigma''_t}{\sigma_t} - \frac{\sigma'_t}{\sigma_t^2}\right)\right]dt \\
&\quad + \left[\left(\frac{1}{\mu_t + \epsilon_t - r}\right)(\mu'_t + 1) + \frac{\sigma'_t}{\sigma_t}\right]dW_t
\end{aligned} \tag{13}$$

The rational expectation equilibrium (REE) can be obtained by comparing the perceived return and the perceived volatility to the realized return and volatility. So, the equilibrium return and volatility can be expressed as a set of equations, which satisfy the following conditions,⁵

$$\begin{aligned}
(\mu_t + \epsilon_t - \frac{1}{2}\sigma_t^2) &= \left[\frac{1}{2}\left(\frac{1}{\mu_t + \epsilon_t - r}\right)[\mu''_t - \left(\frac{1}{\mu_t + \epsilon_t - r}\right)(\mu'_t + 1)] + \frac{1}{2}\left(\frac{\sigma''_t}{\sigma_t} - \frac{\sigma'_t}{\sigma_t^2}\right)\right] \\
\sigma_t &= \left(\frac{1}{\mu_t + \epsilon_t - r}\right)(\mu'_t + 1) + \frac{\sigma'_t}{\sigma_t}
\end{aligned} \tag{14}$$

This implies the REE return and volatility can be represented by a system of nonlinear differential equations, such as

$$\begin{aligned}
g_1(\sigma'_t, \mu'_t, \sigma'_t, \mu'_t, \sigma_t, \mu_t, \epsilon_t, \sigma_\epsilon, r) &= 0 \\
g_2(\sigma'_t, \mu'_t, \sigma_t, \mu_t, \epsilon_t, \sigma_\epsilon, r) &= 0
\end{aligned} \tag{15}$$

From Eq. (15), we can infer the REE return and volatility are complex, nonlinear functions in sentiment, the volatility of sentiment, and an interest rate,

$$\begin{aligned}
\mu_t &= h_1(\epsilon_t, \sigma_\epsilon, r) \\
\sigma_t &= h_2(\epsilon_t, \sigma_\epsilon, r)
\end{aligned} \tag{16}$$

Due to the complex nature of functional forms, it is very difficult to derive the closed-form solutions for the above system, which inhibits us to conduct the comparative static analysis with respect to changes in the sentiment and the interest rate. Therefore, in this paper, we resort to the following empirical model that approximates Eq. (16)

$$\begin{aligned}
\mu_t &= \beta_0 + \beta_1 SI_{t-1} + \beta_2 SV_{t-1} + \beta_3 r_{t-1} + \eta_{1t} \\
\sigma_t &= \alpha_0 + \alpha_1 SI_{t-1} + \alpha_2 SV_{t-1} + \alpha_3 r_{t-1} + \eta_{2t}
\end{aligned} \tag{17}$$

where SI_t denotes the household's sentiment on the housing market, SV_t refers to the volatility of

⁵Note that the perceived price diffusion process is $d\ln P_t = (\mu_t + \epsilon_t - \frac{1}{2}\sigma_t^2)dt + \sigma_t dW_t$.

the sentiment, r_t denotes the interest rate, and η_{it} is a Gaussian error.

We estimate Eq. (17) using the seemingly unrelated regression (SUR) to allow for potential correlation between the housing price and its volatility (Zellner (1962)); that is,

$$\begin{pmatrix} \eta_{1t} \\ \eta_{2t} \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \sigma_{12} \\ \sigma_{12} & 1 \end{pmatrix} \right]$$

This assumption is based on the observation that the housing price change and its volatility are jointly determined from the complex nonlinear relationship in the equilibrium as shown in the theoretical model.

3. Data

We used the monthly home sales price index provided by the KB Real Estate⁶ as a proxy for the real estate price. The monthly mortgage interest rate from the Bank of Korea's Economics Database is used as the interest rate.

News articles related to the real estate markets are obtained from the Naver, Korean-based online search engine. We collected real estate market-related articles released from January 1, 2010 to October 30, 2017 using the following keywords.

- Keywords: real estate, real estate sales, apartment sales

We completed a sentiment analysis on the collected news articles to derive a text-based sentiment index.⁷

We detrended the interest rate variable as it contains a unit root in its dynamic process. Other variables considered in the analysis are determined stationary. The daily sentiment index is initialized as 100 at the beginning period of the sample and then it is standardized. Realized volatility obtained from weekly home sales prices is used for the housing price volatility variable. The monthly sentiment index is obtained by averaging daily sentiment indices; whereas, the monthly volatility of sentiment index is based on the standard deviation of daily sentiment indices in a given month.

Figure 1 plots a series of monthly sentiment indices along with housing price indices. It is interesting to observe the peak in the sentiment index around mid 2015 is followed by the upward trends in housing price indices.

During last several months, the South Korean government introduced several policy measures, including enhanced restrictions on Loan-To-Value (LTV) and Debt-To-Income (DTI) for mortgage loans, to stabilize real estate markets. On the other hand, Figure 2 shows the real estate market sentiment maintained its upward trend before the announcement of the introduction of the second

⁶<http://nland.kbstar.com/quics?page=kbland>

⁷For the details of sentiment analysis, refer to Appendix A.

Table 1: Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
Housing Price Change (%)	0.1889492	0.2223314	-0.1493168	0.9660833
Sentiment Index	8.702111	2.134112	5.604815	14.99755
Sentiment Volatility	3.03648	1.1222	1.688758	7.065116
Interest Rate	-0.6766016	2.564563	-8.701137	8.523044

Source: Authors' calculations, KB Real Estata, Bank of Korea

policy measure, and then turned to flat. This might result from anticipation of the introduction of strong demand suppression policies. A sharp decline in the sentiment index at the end of September is seen as a result of the anticipation of additional regulations scheduled to be released in October.

Figure 3 shows the sentiment volatility has increased significantly in recent months. This observation can be interpreted as reflecting the increased uncertainty caused by the introduction of heightened regulations and foreseeable structural changes in the real estate markets.

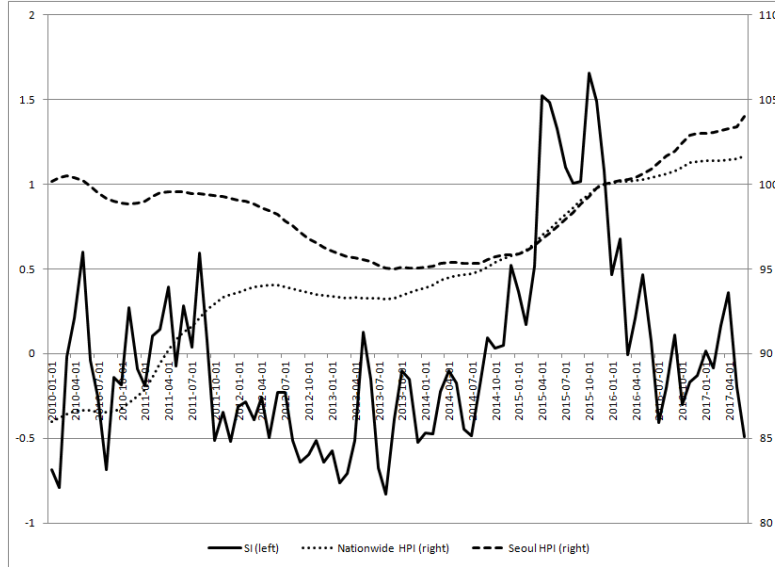


Fig. 1. Monthly Sentiment Index (SI) and Housing Price Indices (HPI)

4. Results

Before we report the main results from SUR, we first investigate simple statistical relationships between the sentiment index and housing price indices. Table 2 shows both past and present real estate sentiment index have a positive correlation with home price indices. This implies improvement in the sentiment index and future housing price increases are closely related. In particular, the correlation between the sentiment index and the nationwide housing price index is relatively larger than the correlation with the Seoul area's housing price index.

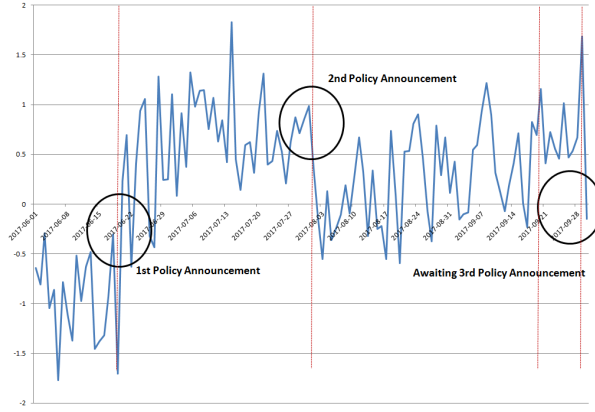


Fig. 2. Daily Sentiment Index

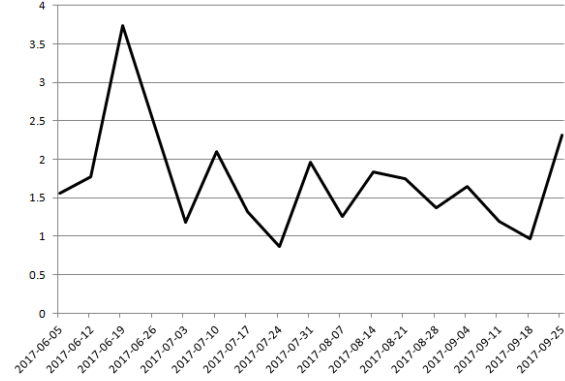


Fig. 3. Sentiment Volatility

Table 2: Correlations between Sentiment Index and Housing Price Index

Variables	SI (t)	SI (t-1)	SI (t-2)	SI (t-3)	SI (t-4)
Nationwide HPI (t)	0.3481	0.381	0.3943	0.4117	0.4372
Seoul HPI (t)	0.1815	0.2415	0.2754	0.3157	0.3705

Note: HPI: housing price index. SI: sentiment index.

Regarding volatility, there is a positive correlation between the volatility of past and current sentiment indices and the volatility of home prices as shown in Table 3. This suggests a sudden change in the real-estate market sentiment is closely related to instability of the future housing market.

Table 3: Correlations between Sentiment Volatility and Housing Price Volatility

Variables	SV (t)	SV (t-1)	SV (t-2)	SV (t-3)	SV (t-4)
Nationwide HPV (t)	0.5081	0.9526	0.9309	0.3007	0.5439
Seoul HPV (t)	-0.0403	0.0840	0.0067	0.2074	0.0574

Note: HPV: housing price volatility. SV: sentiment volatility.

In terms of causality, Table 4 demonstrates the changes in the real estate sentiment index have a statistically significant impact on the change in the nationwide housing price index. On the other hand, the nationwide housing price index in the past does not Granger-cause the sentiment index. This pattern of causality also applies to the case of the Seoul-area housing market as shown in Table 5.

The results of the SUR estimation presented in Table 6 provide additional supporting evidence on the significance of the market sentiment. These results show the improvement in the sentiment index positively affects a month-ahead housing price change for both the nationwide housing market and the Seoul-area housing market. Furthermore, a rise in the real estate sentiment index has been shown to lower the volatility of the nationwide housing price.

Specifically, if the real estate sentiment index increases by one standard deviation, the next

Table 4: Granger-Causality Tests: Nationwide

	lag 1	lag 2	lag 3	lag 4
SI $\rightarrow$ HPI	1.5361 (0.215)	7.4058* (0.025)	9.6981* (0.021)	11.085* (0.026)
HPI $\rightarrow$ SI	0.11484 (0.735)	1.609 (0.447)	1.3767 (0.711)	2.7336 (0.603)

Note: * $p < 0.1$. Values in parentheses denote p-values. HPI: housing price index. SI: sentiment index.

Table 5: Granger-Causality Tests: Seoul

	lag 1	lag 2	lag 3	lag 4
SI $\rightarrow$ HPI	1.7842 (0.182)	5.3614* (0.069)	7.7305* (0.052)	6.5134 (0.164)
HPI $\rightarrow$ SI	1.7622 (0.184)	3.2704 (0.195)	3.4642 (0.325)	7.6155 (0.107)

Note: * $p < 0.1$. Values in parentheses denote p-values. HPI: housing price index. SI: sentiment index.

month’s nationwide house price index rises by 0.16 percentage point and volatility drops by about 0.0109 percentage point. On the other hand, a rise in the sentiment index predicts a positive price change for the Seoul area, but not for volatility. A one standard deviation increase in the sentiment index predicts a positive increase in the Seoul-area housing price index by 0.298 percentage point.

However, the sentiment volatility is negatively related to the Seoul-area’s housing price change. Our findings suggest the rise in volatility of the sentiment index led to a 0.256 percentage point drop in the one-month-ahead Seoul-area housing price index.

Correlations between price change and volatility are shown different between the nationwide market and the Seoul-area market. For the nationwide price index, there is a statistically positive correlation at the magnitude of 0.2073; whereas, no significant correlation exists for the Seoul-area price index. This finding suggests the instability of the housing market is likely to intensify when the market is on the boom side of a cycle.

This partial asymmetry in the results is not surprising given the fact news articles used in the analysis include not only the Seoul-area market but also other South Korean areas. Therefore, we conjecture the sentiment index we constructed in this paper seems more relevant for nationwide housing market variables.

5. Concluding Remarks

Even though this paper is a preliminary study on the South Korea’s housing market using text-mining analytics, it offers important policy implications. Korean households have accumulated sizeable household debts to finance their housing purchases during recent years. The normalization of monetary policies in the advanced economies and anticipated benchmark rate hikes by South Korea’s central bank will pose significant threats to the fiscal soundness of Korean households.

Table 6: Seemingly Unrelated Regression Estimates

VARIABLES	Nationwide		Seoul	
	HPC	HPV	HPC	HPV
SI (t-1)	0.160*** (0.0456)	-0.0109* (0.00602)	0.298*** (0.0393)	2.30E-05 (0.0000575)
SV (t-1)	-0.0305 (0.0872)	0.011 (0.0115)	-0.256*** (0.0751)	7.60E-05 (0.00011)
Mortgage Rate (t-1)	0.0143 (0.00877)	-0.000741 (0.00116)	0.00808 (0.00756)	1.60E-05 (0.0000111)
Constant	0.221*** (0.0727)	0.0347*** (0.0096)	0.245*** (0.0627)	0.000352*** (0.0000917)
Correlation		0.2073* (3.739)		-0.0169 (0.025)
Observations	87	87	87	87
R-squared	0.189	0.048	0.433	0.037

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Values in parentheses denote p-values. Breusch-Pagan test statistics of independence are shown in the parentheses. HPC: housing price change. HPV: housing price volatility SI: sentiment index. SV: sentiment volatility.

It is highly probable the current situation can significantly worsen household confidence in the housing market, which may result in adverse macroeconomics consequences, e.g., sharp decline in household consumption and an elevated systemic risk on the South Korean banking system. To stabilize the housing market in this turbulent environment, policy-makers should have specific operational tools to manage the household sentiment properly. In addition, provided the housing price volatility is positively associated with sentiment volatility, the government must be consistent in formulating housing market-related policies to prevent dramatic swings in the household sentiment. Moreover, the existing market surveillance system employed by the financial regulation authority should be complemented with qualitative information from various sources.

References

- Antweiler, Werner and Murray Z. Frank. 2004. Is all that talk just noise? The information content of Internet stock market message boards, *The Journal of Finance*, 59(3): 1259–1294.
- Baker, Malcolm and Jeffrey Wurgler. 2007. Investor sentiment and cross-section of stock returns, *The Journal of Finance*, 61(4): 1645–1680.
- Bholat, David, Stephen Hansen, Pedro Santos, and Cheryl Schonhardt-Bailey. 2015. Text-mining for central banks: handbook, *Centre for Central Banking Studies*, 33: 1–19.
- Danielsson, Jon, Hyun Song Shin, and Jean-Pierre Zigrand. 2010. Risk appetite and endogenous risk, AXA Working Paper Series No. 2.
- Doran, J.S., D. R. Peterson, and S.M. Price. 2010. Earnings conference call content and stock price: The case of REITs, *Journal of Real Estate Finance and Economics*, 45(2): 402–434.
- Garcia, Diego. 2013. Sentiment during recessions, *Journal of Finance*, 68(3): 1267–1300.
- Kothari, S.P., X. Li, and J.E. Short. 2009. The effect of disclosures by management, analysts, and business press on cost of capital, return volatility, and analyst forecasts: A study using content analysis, *The Accounting Review*, 50(2): 1639–1670.
- Li, F. 2010. The information content of forward-looking statements in corporate filings-A naive Bayesian machine learning algorithm approach, *Journal of Accounting Research*, 48: 1049–1102.
- Nymand-Andersen, Per. 2016. Big data: The hunt for timely insights and decision certainty, *IFC Working Papers No. 14*.
- Li, F. 2010. Momentum traders in the housing market: Survey evidence and a search model, *American Economic Review: Papers & Proceedings*, 99(2): 406–411.
- Pyo, Dong-Jin and Jungho Kim. 2017. News media sentiment and asset prices: Text-mining approach, *KIF Working Paper 2017-09*.
- Soo, Cindy K. 2015. Quantifying animal spirits: News media and sentiment in the housing market, *Ross School of Business Working Paper No. 1200*.
- Shiller, Robert. 2005. Irrational Exuberance, *Princeton University Press*, New Jersey, USA
- Tetlock, Paul C. 2007. Giving content to investor sentiment: The role of media in the stock market, *Journal of Finance*, 62(3): 1139–1168.
- Tetlock, Paul C., M. Saar-Tsechansky, and S. Macskassy. 2008. More than words: Quantifying language to measure firm fundamentals, *Journal of Finance*, 63(3): 1437–1467.

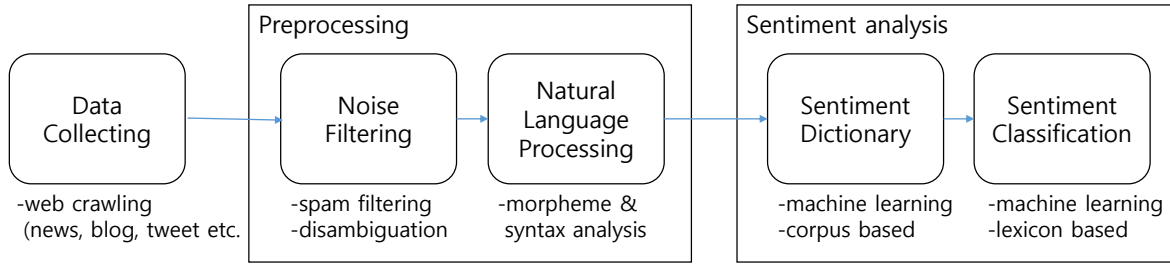
Wu, Lynn and Erik Brynjolfsson. 2013. The Future of Prediction: How Google Searches Foreshadow Housing Prices and Sales, mimeo.

Zellner, Arnold. 1962. An efficient method of estimating seemingly unrelated regressions and tests for aggregation bias, *Journal of the American Statistical Association*, 57: 348–368.

Appendix A. Sentiment Analysis

Sentiment analysis⁸ refers to a set of computational techniques that aim to extract sentimental information from unstructured texts, such as news articles, comments, and social network services. In this paper, we build a collective emotional state of market participants on the real estate market from online newspaper articles using sentiment analysis.

The sentiment analysis process usually consists of three stages: (1) data collection and processing, (2) development of emotional speech dictionary, and (3) sensitivity classification.



Source: Pyo and Kim (2017)

Fig. 4. The Process of Sentiment Analysis

- **Data Collection:** Collect text data online using web crawler, etc.
- **Preprocessing:** As a preprocessing step, data are converted into an easy-to-analyze form through noise data removal and Natural Language Processing (NLP).
- **Build a Sentiment Dictionary:** Construct a dictionary⁹ with vocabulary (phrases and words) containing sentiment information. In this state, we define a few sentiment words in advance and expand the sentiment dictionary by adding lexicons closely related to them.¹⁰
- **Sentiment Classification:** Based on the lexicons of the sentiment dictionary, the sentiment of the text is classified as positive or negative. Sentiment classification methods are divided

⁸This section is reproduced from the Section 3 of Pyo and Kim (2017). More detailed description on sentiment analysis and related examples can be found here.

⁹This refers to a set consisting of emotional and emotional expressions. Emotion information is expressed as a real number in the range [-1, 1]. A negative (positive) value corresponds to the negative (positive) opinion. As closer to both extreme, sentiment intensity becomes stronger)

¹⁰Based on co-occurrence information for sentiment lexicons, a sentiment dictionary is constructed using SO-PMI (Sentiment Orientation Point-wise Mutual Information), which obtains the direction and degree of sentiment of a specific lexicon.

$$\text{SO-PMI}(w) = \sum_{i=1}^{N_p} \text{PMI}(w, p_i) - \sum_{j=1}^{N_n} \text{PMI}(w, n_j) \quad (18)$$

where

$$\text{PMI}(w_1, w_2) = \log \frac{P(w_1, w_2)}{P(w_1)P(w_2)} \quad (19)$$

into Lexicon-Based and Model-Based methods. The Lexicon-Based methodology determines the orientation of the text by combining the information such as the appearance frequency of sentiment words in the text or the sentiment intensity. On the other hand, the Model-Based method constructs a classification model using classified training data, and classifies the sentiment of the text.¹¹

$$so_i = \begin{cases} positive & \text{if } s_i > 0 \\ neutral & \text{if } s_i = 0 \\ negative & \text{otherwise} \end{cases} \quad (20)$$

where $s_i = f_p(x_i) - f_n(x_i)$, and $f_p(x_i)$ ($f_n(x_i)$) denotes the frequency of positive (negative) lexicons in the text x_i

¹¹Considering it is difficult to predefine training data with a large number of stakeholders related to the real estate market, we use the Lexicon-Based method in this study. In recent years, the Ensemble Learning Method, which synthesizes the results of multiple models to improve the accuracy of the analysis by supplementing the limits of a single model, is also used.